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Software Requirements Specification (SRS)

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PROJECT CHARTER

1.1 PROJECT OBJECTIVE

ID	Objective	Priority
OBJ-01	To digitize patient registration and eliminate duplicate records through a centralized database	High
OBJ-02	To automate appointment booking with real-time doctor schedule validation and double-booking prevention	High
OBJ-03	To provide role-based dashboards for Admin, Patient, Doctor, and Barangay Staff	High
OBJ-04	To generate automated LGU health reports with service utilization and demographic analytics	Medium
OBJ-05	To replace manual logbooks and unstructured communication (Viber, notebooks) with a digital system	High



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This project aims to modernize the clinic by replacing messy handwritten notebooks and scattered chat groups with a single, secure digital system. The new software will wipe out confusing duplicate patient files and automatically block schedule conflicts so doctors are never double-booked. It provides custom, easy-to-use digital dashboards tailored specifically for what patients, staff, doctors, and administrators need to see. Finally, the system takes the headache out of paperwork by instantly generating clean health reports for the local government unit.

1.2 SMART Success Criteria

Criteria	SMART Statement	Target	Measurement Method
Specific	Patients shall book an appointment from login to confirmation within 2 minutes	2 minutes	Time tracking during user testing
Measurable	The system shall prevent 100% of double booking attempts through database triggers	100%	Error log review and testing
Achievable	The system shall provide fully functional dashboards for all 4 user roles	4 roles	Functional testing and user acceptance
Relevant	LGU report generation time shall be reduced from 2 hours (manual) to under 5 minutes	5 minutes	Before/after comparison with manual process



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Time-bound	The complete system shall be deployed and tested within 7 weeks from project start	7 weeks	Project schedule tracking
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1.3 High-Level Risks

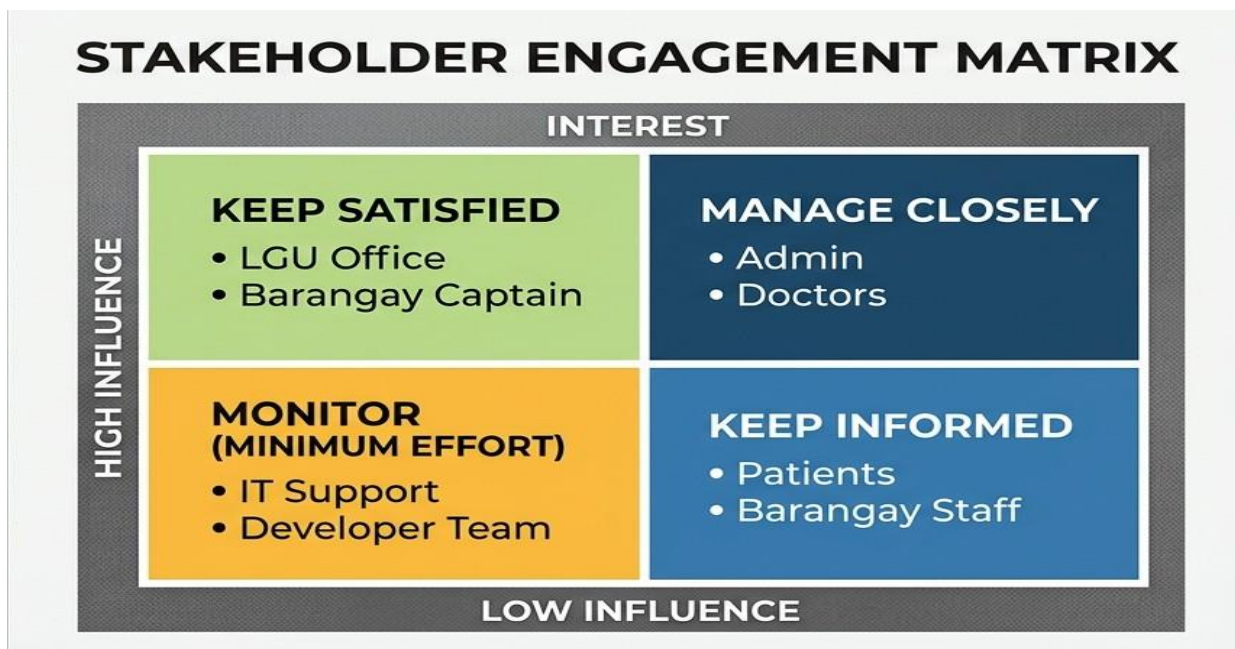
ID	Risk Description	Probability	Impact	Mitigation Strategy
R-01	Staff resistance to adopting new digital system	Medium	High	Provide training sessions; demonstrate benefits over manual logbook
R-02	Hardware failure (computer breakdown)	Low	High	Implement regular database backups, have backup computer available
R-03	Power outage during clinic hours	Medium	Medium	Use UPS for server computer



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R-04	Data privacy compliance issues	Low	High	Implement role-based access; obtain patient consent forms
R-05	Learning curve for non-tech savvy users	Medium	Medium	Create simple user manual; provide in-app tooltips
R-06	Database corruption or data loss	Low	High	Weekly automated backups; test restore procedures

2 Stakeholder Analysis





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This chart serves as a simple guide for managing project relationships by grouping people into four boxes based on their level of interest and decision-making power. The clinic administrators and doctors are placed in the top-right box because they will use the system daily, meaning the project team must work hand-in-hand with them. The local government unit (LGU) office and the barangay captain sit in the top-left box because they hold high official authority, requiring the system to satisfy their specific reporting needs. Lastly, the general clinic staff and patients are placed in the bottom sections because they primarily need to be kept informed about system updates rather than directing the project's development.

2.1 Negotiation Log

ID	Stakeholder	Request/Concern	Resolution	Status
N-01	Admin	"We need to prevent double booking completely"	Implemented database trigger that checks for existing appointments before insert	Resolved
N-02	Doctors	"We need to add medical notes to patient records"	Added "Add Medical Notes" button for confirmed appointments	Resolved



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N-03	Admin	"Patient records should not have duplicates"	Added UNIQUE constraint on contact_no; duplicate checking on name	Resolved
N-04	LGU Office	"Reports should be easy to read and export"	Added bar charts, CSV export, print functionality	Resolved
N-05	Admin	"Doctors have different schedules"	Implemented doctor-specific schedule validation in triggers	Resolved
N-06	Staff	"We need to search residents quickly"	Added real-time search functionality on patient tables	Resolved
N-07	Admin	"Only authorized users should access data"	Implemented session-based authentication with role checks	Resolved

3. REQUIREMENTS ELICITATION

3.1 Informal System Artifacts Identified



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Artifact	Description	Problem Identified
Handwritten Logbook	Physical notebook for recording appointments	Pages get lost or torn; no search functionality; cannot prevent double booking
Viber Group Chat	Mobile messaging for schedule changes	Messages get buried; no organized tracking; no confirmation of receipt
Small Notebook	Pocket notebook for patient names	Names misspelled or forgotten; no follow-up mechanism

3.2 Requirements Elicitation Table

Category	Definition	Example from Your Project (Raw Quote)
Business Rule	A strict policy, regulation, or legal constraint that must always be followed	<i>"Patients must be residents of the barangay. Non-residents may still be accommodated but will not be prioritized."</i>



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	and enforced by the system.	
Functional Need	A specific action, task, or feature the system must perform to support user goals.	<i>"We need a system that can book appointments and automatically check if the doctor is available on that day and time."</i>
Functional Need	A specific system action to prevent data inconsistency.	<i>"The system should prevent me from adding the same patient twice, especially if they have the same name and contact number."</i>
Functional Need	A specific system action to support reporting.	<i>"The system must generate a summary of appointments by service type so we know how many patients used General Checkup, Prenatal, and Pediatric."</i>
Quality Attribute (NFR)	A non-functional requirement describing how well the system should perform.	<i>*"The system should generate reports quickly because we submit them to the LGU every month. It should not take more than 2-5 seconds."*</i>
Quality Attribute (NFR)	A security and privacy requirement.	<i>"Resident health data must be protected from unauthorized access. Only authorized staff, doctors, and the patients themselves should see medical records."</i>



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Assumption	A belief about user cooperation.	<i>"We assume that patients will provide accurate contact numbers so we can reach them for appointment confirmation."</i>
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4. SPECIFIC REQUIREMENTS

4.1 Functional Requirements (FR) - MoSCoW Prioritized

ID	Requirement (Using "Shall" syntax)	Priority
FR-01	The system shall prevent double booking by validating that no two appointments exist for the same doctor at the same date and time	Must
FR-02	The system shall validate doctor schedules and reject appointments outside their specific working days and hours	Must
FR-03	The system shall store patient information in a centralized database and eliminate	Must



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	duplicate records using unique contact numbers	
FR-04	The system shall generate summary reports using automated queries and views for efficient data retrieval	Should
FR-05	The system shall validate all appointment inputs (date, time, doctor, service, patient) and reject invalid entries before saving	Must
FR-06	The system shall enforce doctor-service specialization rules where each doctor can only perform their designated medical service	Must
FR-07	The system shall allow patients to view their own health records directly from their patient dashboard	Should
FR-08	The system shall perform duplicate checking on patient name and contact number before inserting new records	Must
FR-09	The system shall provide service utilization reports showing counts for General Checkup, Prenatal, and Pediatric	Should



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FR-10	The system shall allow Admin to update appointment status (Pending, Confirmed, Completed, Cancelled)	Must
FR-11	The system shall automatically create a health record entry when appointment status changes to "Completed"	Must
FR-12	The system shall allow doctors to add medical notes to confirmed appointments	Should
FR-13	The system shall provide search functionality for patients by name, address, or contact number	Should
FR-14	The system shall authenticate users with role-based access control	Must

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FR-03	The system shall store patient information in a centralized database and eliminate duplicate records using unique contact numbers	Must
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FR-08	The system shall perform duplicate checking on patient name and contact number before inserting new records	Must
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4.2 Non-Functional Requirements (NFR) - MoSCoW Prioritized

ID	Requirement (Using "Shall" syntax)	Priority
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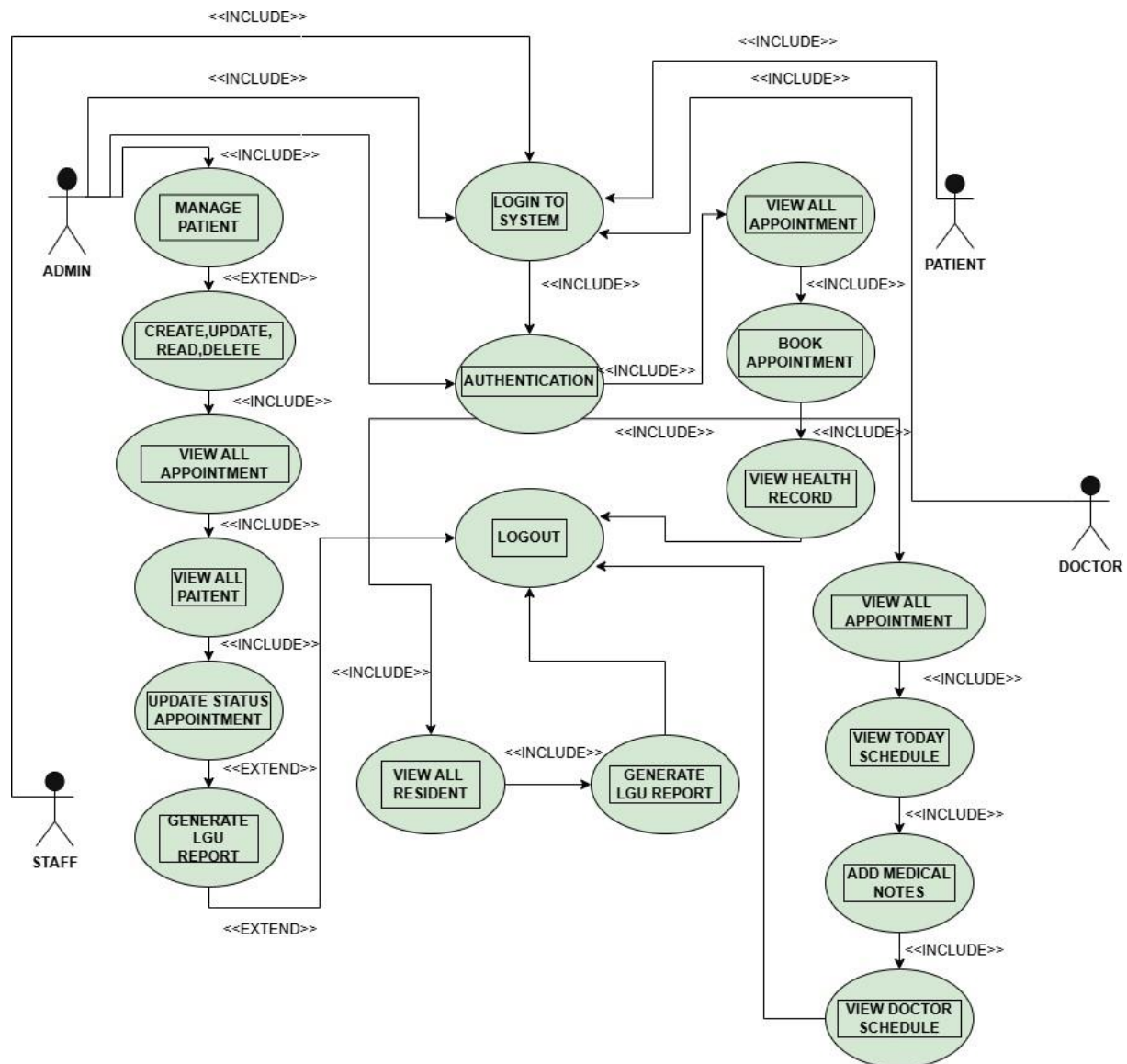


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NFR-01	The system shall return report query results within 2-5 seconds under normal operating conditions	Should
NFR-02	The system shall implement role-based access control where Admin, Doctor, Patient, and Staff have different permission levels	Must
NFR-03	The system shall ensure transactional integrity using ACID compliant database transactions	Must
NFR-04	The system shall provide intuitive user interfaces with clear buttons, labels, and form validations	Should
NFR-05	The system shall enforce patient-specific data isolation where patients can only access their own health records	Must



LOGICAL USE CASE MODELS





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Description:

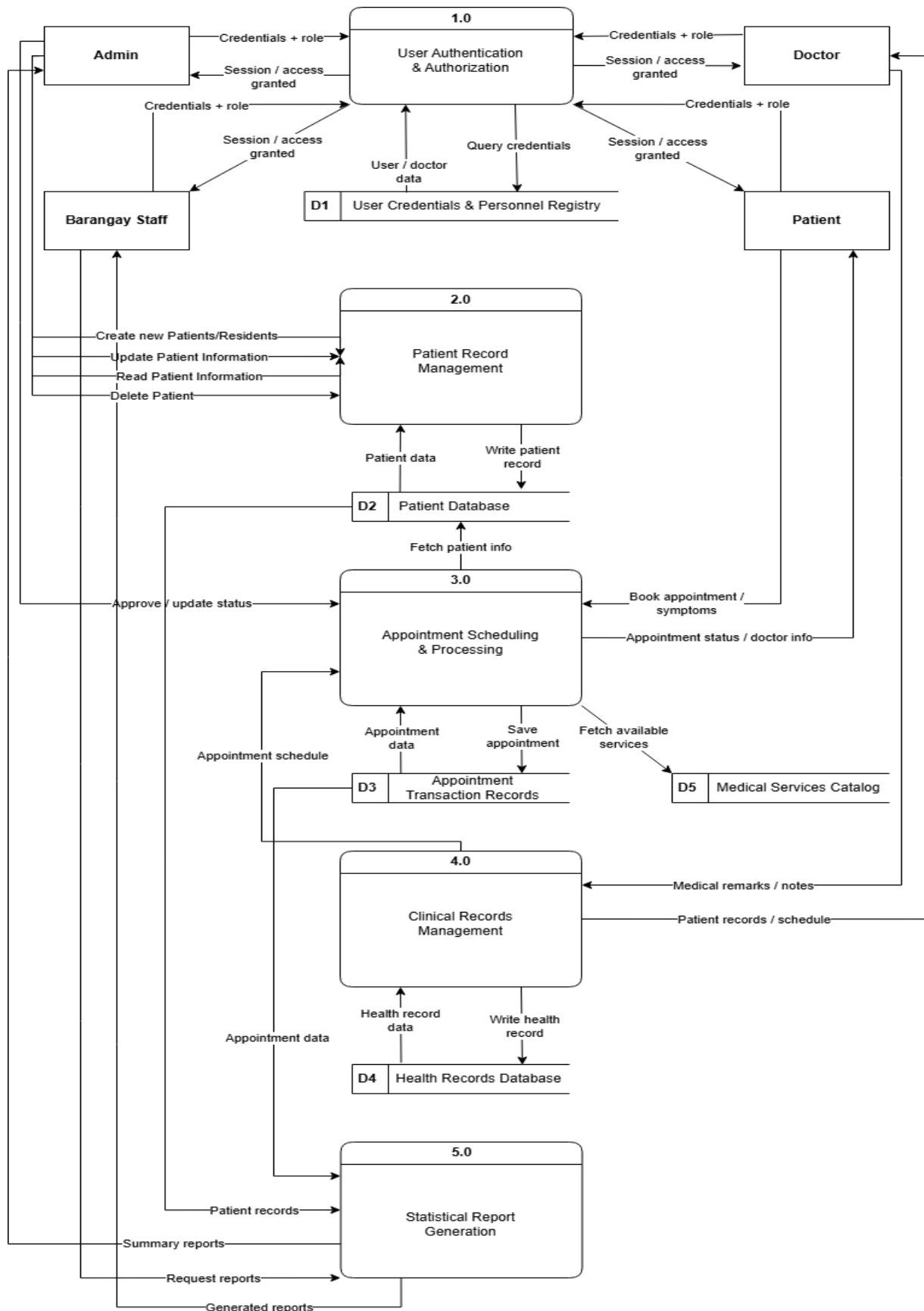
The Use Case Diagram of the Barangay Health Clinic System shows the interactions between the Admin, Barangay Staff, Patient, and Doctor within the system. It illustrates the main functions each user can access, such as managing patient records, booking appointments, viewing schedules, adding medical notes, generating reports, and monitoring health records. The diagram also includes shared system functions like login, authentication, and logout.

System Boundary

LEVEL 1



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Description:

The Level 1 Data Flow Diagram (DFD) of the Barangay Health Clinic System shows the major processes and how data moves within the system. It breaks down the main system into five core processes: User Authentication and Authorization, Patient Record Management, Appointment Management, Medical Records Management, and Report Generation.

The diagram illustrates how the Admin, Barangay Staff, Doctor, and Patient interact with the system, while data is stored and retrieved from the different databases such as Users, Patients, Appointments, Health Records, and Services. It also shows how the system manages user access, patient records, appointment scheduling, medical information, and report generation.